

Lecture #5

Content to be covered

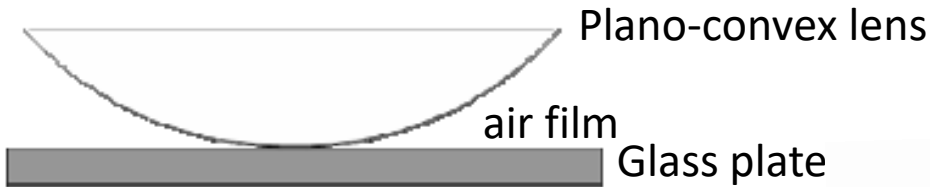
- ❑ Newton's Ring (with reflected light)
 - ❖ Ring radii
 - ❖ Relation between film thickness and ring radius
- ❑ Applications of Newton's Ring experiment
 - ❖ Wavelength determination
 - ❖ Refractive index determination
 - ❖ Optical flatness of a glass plate
- ❑ Case study (displacement of lens)

Resources to be consulted

- ❑ OPTICS by Ajay Ghatak

Newton's Ring

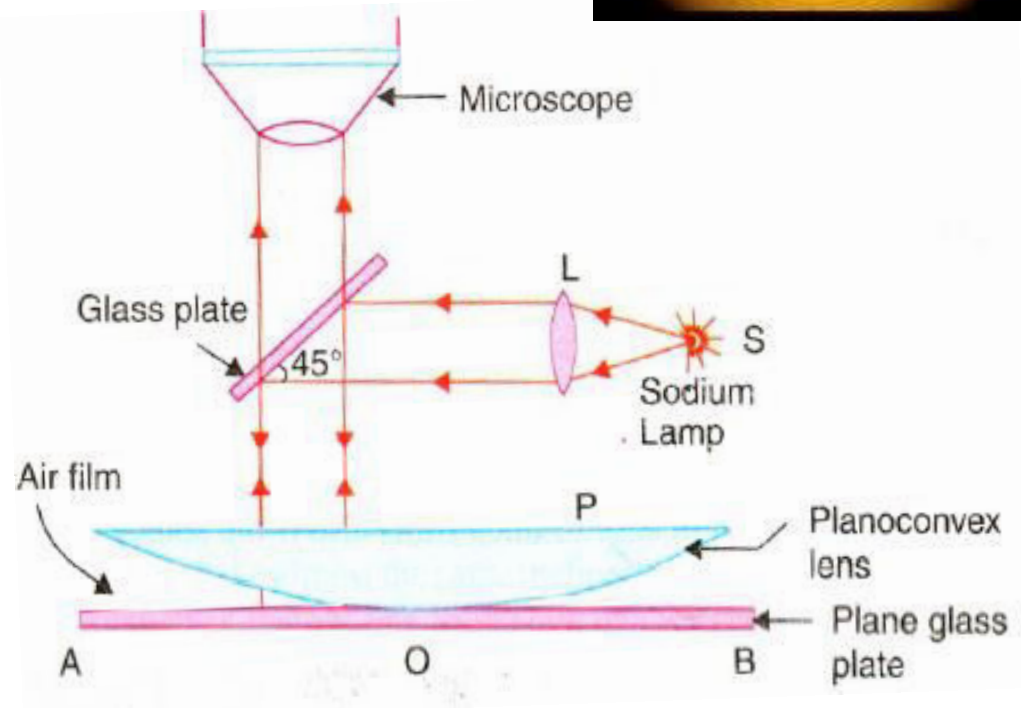
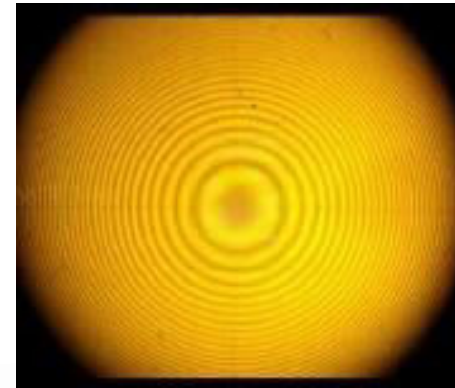
When a plano-convex lens of large radius of curvature is placed with its convex surface in contact with a plane glass plate, an air film is formed between the lower surface of lens and the upper surface of the plate.



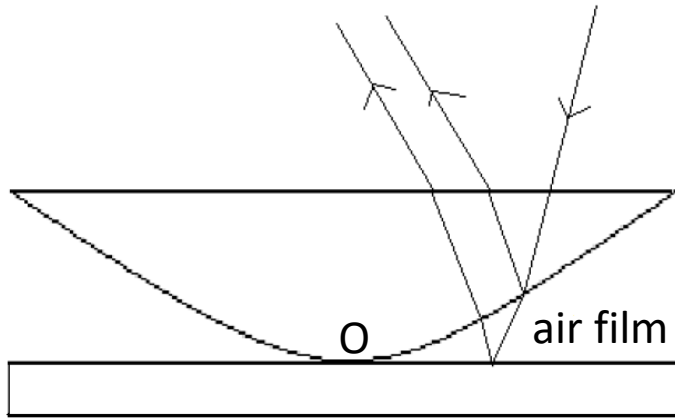
Thickness of air film is zero at point of contact.

If monochromatic light is allowed to fall normally on this film, a system of alternate bright and dark concentric Newton's rings are formed as a result of interference between the light waves reflected from the upper and lower surface of air film.

Interference occurs between the light reflected from the lower surface of plano-convex lens and the upper surface of the glass plate.



Newton's Ring with reflected light



From reflected light

(i) For bright rings: $2\mu t \cos r = (2n + 1) \frac{\lambda}{2}$

For air film, $\mu = 1$ and for near normal Incidence, r is very small and hence $\cos r \approx 1$

$$2t = (2n + 1) \frac{\lambda}{2}$$

$n = 0, 1, 2, \dots$

(ii) For dark rings: $2\mu t \cos r = n\lambda$

$$2t = n\lambda$$

Why do you get circular rings in Newton's ring experiment?

Since the convex side of the lens is a spherical surface, thickness of the air film will be constant over a circle (whose centre will be at O) and we will obtain concentric dark and bright rings.

Each ring will be locus of all such points where thickness is same.

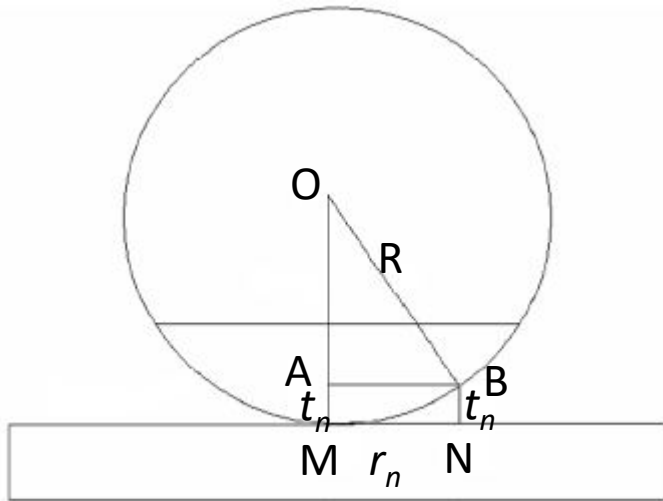
From transmitted light

(i) For bright rings: $2t = n\lambda$

(ii) For dark rings: $2t = (2n + 1) \frac{\lambda}{2}$

in Newton's rings for transmitted rays, the central ring will be bright.

Newton's Ring with reflected light



R = radius of curvature of lens

t = thickness of air film at a distance $AB = r_n$

$OA = R - t$

From $\triangle OAB$: $OB^2 = OA^2 + AB^2$

$$R^2 = (R - t_n)^2 + r_n^2$$

$$\Rightarrow r_n^2 = 2Rt_n - t_n^2 \quad R \gg t_n \text{ then } r_n^2 = 2Rt_n$$

Now the condition for **bright rings from Reflection**

$$2t = (2n+1)\frac{\lambda}{2} = \frac{r_n^2}{R} \Rightarrow D_n = 2\sqrt{\frac{(2n+1)\lambda R}{2}} \quad n = 0, 1, 2, \dots$$

similarly for **dark rings**: $2t = n\lambda = \frac{r_n^2}{R} \Rightarrow D_n = 2\sqrt{n\lambda R}$

For $n = 0$ $r_{\text{bright}} = \sqrt{\lambda R/2}$ and $r_{\text{dark}} = 0$

Thus **center would be dark** of the fringe pattern and alternately dark and bright rings will be produced.

Newton's Ring with reflected light

The diameter of the dark rings is $D_n = 2\sqrt{n\lambda R}$ $n = 1, 2, 3, \dots$

The spacing between second and third dark rings is smaller than the spacing between the first and second one.

$$D_1 = 2\sqrt{\lambda R}$$

$$D_2 = 2(1.41)\sqrt{\lambda R}$$

$$D_3 = 2(1.73)\sqrt{\lambda R}$$

$$D_4 = 2(2)\sqrt{\lambda R}$$

$$D_5 = 2(2.24)\sqrt{\lambda R}$$



Although the diameters of the dark rings are proportional to the square root of natural numbers, but the diameter of the ring does not increase in the same proportion as the order of the ring. Hence, the rings get closer and closer as the order of rings i.e. 'n' increases.

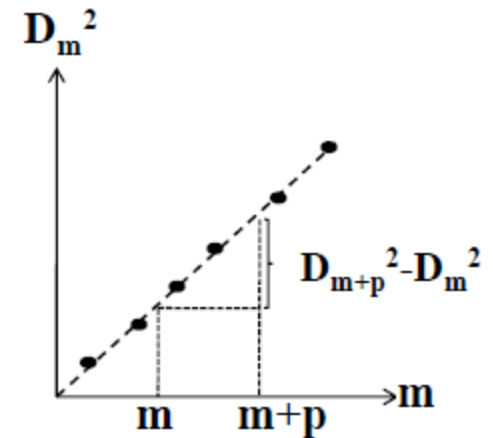
Newton's rings : Wavelength determination

Diameter of the n^{th} dark ring (r_n) is given by $D_n^2 = 4n\lambda R$

Similarly for $(n+m)^{\text{th}}$ dark band : $D_{n+m}^2 = 4(n+m)\lambda R$

$$D_{n+m}^2 - D_n^2 = 4(n+m)\lambda R - 4n\lambda R$$

$$\text{or } \lambda = \frac{D_{n+m}^2 - D_n^2}{4mR}$$



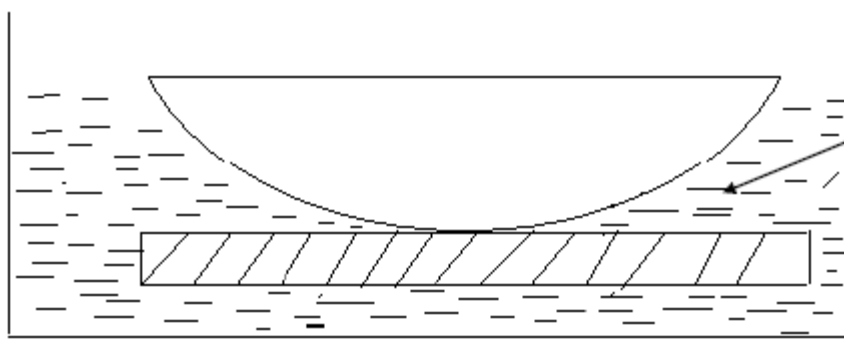
The slope of the straight line gives the value of $4\lambda R$.

Suppose diameter of 5^{th} and 15^{th} ring are determined then, $m = 15 - 5 = 10$

$$\lambda = \frac{D_{15}^2 - D_5^2}{4 \times 10 \times R}$$

Radius of curvature can be accurately measured with the help of a spherometer and therefore by measuring the diameter of dark or bright ring one can experimentally determine the wavelength.

Newton's Ring : Refractive index determination



Liquid of refractive index μ

for near normal incidence and $\mu_g > \mu$

First, perform the experiment in air

Diameter of the dark rings $D_n^2 = 4n\lambda R$ $D_{n+m}^2 = 4(n+m)\lambda R$ $D_{n+m}^2 - D_n^2 = 4m\lambda R$ (1)

Now, repeat the experiment in liquid

The path difference for dark rings will be given by $2\mu t = n\lambda$ but $t = \frac{r_n^2}{2R}$

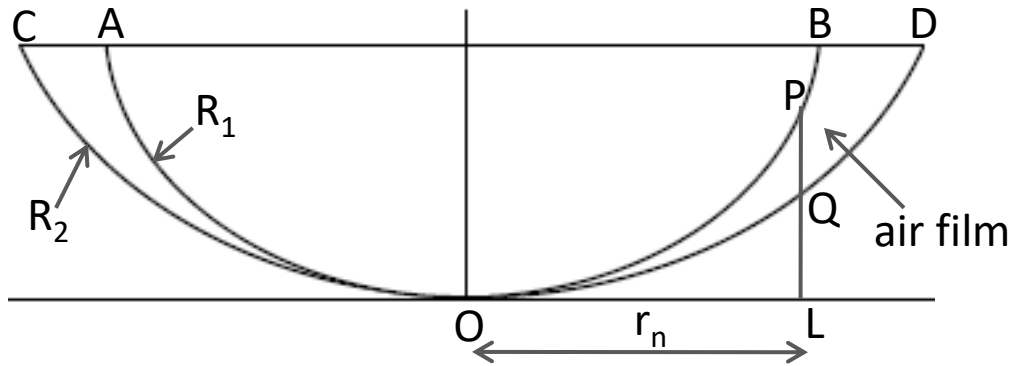
$$D_n^2 = 4n\lambda R / \mu$$

$$\Rightarrow D_{n+m}^2 - D_n^2 = 4m\lambda R / \mu \text{(2)}$$

now take the difference between (1) and (2)

$$\mu = \frac{D_{n+m}^2 - D_n^2}{D_{n+m}^2 - D_n^2}$$

Newton's Ring between two curved surfaces



- Convex and concave surfaces
- Effective air film thickness decreases

PQ be the thickness (say t) of air film at point P, thus $t = PQ = PL - QL$

$$PL = \frac{r_n^2}{2R_1} \text{ and } QL = \frac{r_n^2}{2R_2} \Rightarrow t = \frac{r_n^2}{2R_1} - \frac{r_n^2}{2R_2} = \frac{r_n^2}{2} \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

The condition for **bright rings** for air film

$$2t = (2n+1) \frac{\lambda}{2} \quad n = 0, 1, 2, \dots$$

$$\Rightarrow r_n^2 \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = (2n+1) \frac{\lambda}{2}$$

$$\text{or } D_n^2 = \frac{2(2n+1)\lambda}{\left[\frac{1}{R_1} - \frac{1}{R_2} \right]}$$

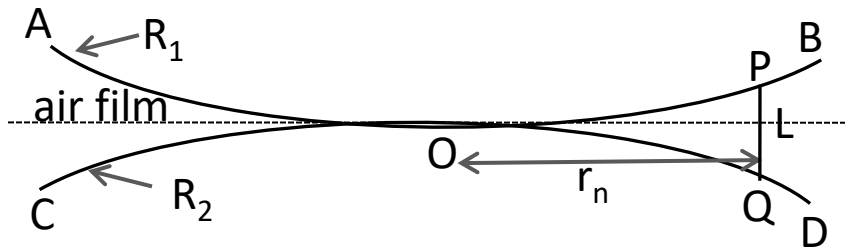
The condition for **dark rings** for air film

$$2t = n\lambda \quad n = 0, 1, 2, \dots$$

$$\Rightarrow r_n^2 \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = n\lambda$$

$$\text{or } D_n^2 = \frac{4n\lambda}{\left[\frac{1}{R_1} - \frac{1}{R_2} \right]}$$

Newton's Ring between two curved surfaces



- Both convex surfaces
- Effective air film thickness increases

PQ be the thickness (say t) of air film from point O, thus $t = PQ = PL + QL$

$$PL = \frac{r_n^2}{2R_1} \text{ and } QL = \frac{r_n^2}{2R_2} \Rightarrow t = \frac{r_n^2}{2R_1} + \frac{r_n^2}{2R_2} = \frac{r_n^2}{2} \left[\frac{1}{R_1} + \frac{1}{R_2} \right]$$

The condition for **bright rings** for air film

$$2t = (2n+1) \frac{\lambda}{2} \quad n = 0, 1, 2, \dots$$

$$\Rightarrow r_n^2 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = (2n+1) \frac{\lambda}{2}$$

$$\text{or } D_n^2 = \frac{2(2n+1)\lambda}{\left[\frac{1}{R_1} + \frac{1}{R_2} \right]}$$

The condition for **dark rings** for air film

$$2t = n\lambda \quad n = 0, 1, 2, \dots$$

$$\Rightarrow r_n^2 \left[\frac{1}{R_1} + \frac{1}{R_2} \right] = n\lambda$$

$$\text{or } D_n^2 = \frac{4n\lambda}{\left[\frac{1}{R_1} + \frac{1}{R_2} \right]}$$

Determination of optical flatness of a glass plate

Consider a glass surface placed on another surface whose flatness is known. If a monochromatic light beam is allowed to fall on this combination and the reflected light is viewed by a microscope, then, in general, dark and bright rings will be seen.

$$\text{For bright rings: } 2t = (2n + 1) \frac{\lambda}{2}$$

$$n = 0, 1, 2, \dots$$

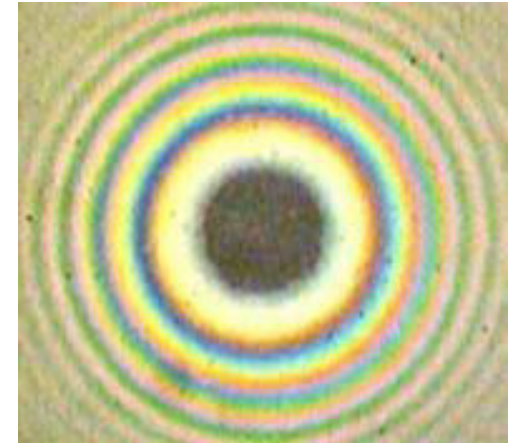
$$\text{For dark rings: } 2t = n\lambda$$

Consequently, by measuring the distance between consecutive dark and bright rings, one can calculate the optical flatness of a glass surface.

Newton's Ring with white light

If the monochromatic source is replaced by the white light few colored rings are seen around dark centre later illumination is seen in the field of view..

However, if we put a red filter in front of the eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of the eye, the fringe pattern corresponding to the green color will appear.

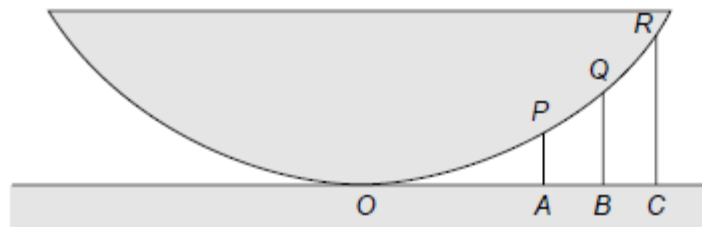


Newton's Ring: Can we make centre bright in reflected rays?

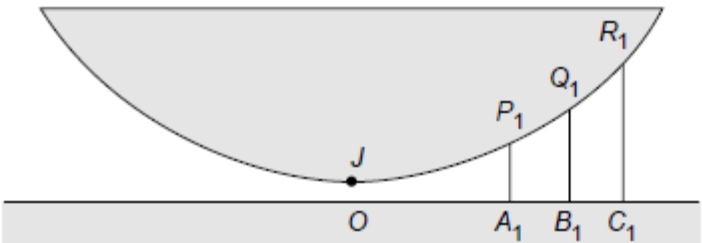
- ❑ By using a liquid film with refractive index μ_{Liquid} with condition $\mu_{\text{Convex lens}} < \mu_{\text{Liquid}} < \mu_{\text{Glass plate}}$. For example: crown glass=1.45, flint glass=1.63 and liquid with $1.45 < \mu < 1.63$.
- ❑ By lifting convex lens upward with a distance $\lambda/4$. because $2t=n\lambda$ (for dark rings)
 $2[t + (\lambda/4)] = n\lambda + \lambda/2 = (2n+1)\lambda/2$ (bright)

Newton's Ring : Vertical shifting of lens (Case1)

Consider the formation of Newton's rings by monochromatic light. Assume the point of contact to be perfect. Now slowly raise the lens vertically above the plate. As the lens moves gradually away from the plate, how ring pattern will change.



$$\begin{aligned} PA &= \lambda/2 \\ QB &= \lambda \\ RC &= 3\lambda/2 \end{aligned}$$



$$\begin{aligned} JO &= \lambda/4 \\ P_1A_1 &= \lambda/2 \\ Q_1B_1 &= \lambda \\ R_1C_1 &= 3\lambda/2 \end{aligned}$$

Since the point of contact is perfect, the central spot will be dark, the first dark ring will form at P where $PA = \lambda/2$.

If we now raise the lens by $\lambda/4$, then $2t$ corresponding to the central spot would be $\lambda/2$ and instead of the dark spot at the center we will now have a bright spot.

If the lens is further moved by $\lambda/4$, then the first dark ring collapses to the center and the central spot will be dark.

Thus, as the lens is moved upward, the rings collapse to the center.

Newton's Ring : Vertical shifting of lens (Case2)

Consider the formation of Newton's rings when two closely spaced wavelengths are present; for example, the D1 and D2 lines of sodium ($\lambda_1 = 5890 \text{ \AA}$ and $\lambda_2 = 5896 \text{ \AA}$). What will be the effect of the presence of these two wavelengths as the lens is gradually moved away from the plate?

For $\lambda = 5.890 \times 10^{-5} \text{ cm}$,

$$\begin{aligned}\text{Radius of ninth bright ring} &= \sqrt{\left(9 + \frac{1}{2}\right)\lambda R} \\ &= \sqrt{9.5 \times 5.890 \times 10^{-5} \times 100} \\ &= 0.236548 \text{ cm}\end{aligned}$$

$$\begin{aligned}\text{Radius of ninth dark ring} &= \sqrt{9\lambda R} \\ &= 0.230239 \text{ cm}\end{aligned}$$

Similarly, for $\lambda = 5.896 \times 10^{-5} \text{ cm}$,

$$\begin{aligned}\text{Radius of ninth bright ring} &= \sqrt{9.5 \times 5.896 \times 10^{-5} \times 100} \\ &= 0.236669 \text{ cm}\end{aligned}$$

and

$$\begin{aligned}\text{Radius of ninth dark ring} &= \sqrt{9 \times 5.896 \times 10^{-5} \times 100} \\ &= 0.230356 \text{ cm}\end{aligned}$$

We will first assume that the lens is in contact with the plane glass plate. Since the **two wavelengths are very close**, the **bright and dark rings of λ_1 superpose on the bright and dark rings of λ_2** , respectively.

Thus the rings almost exactly superpose on each other.

Newton's Ring : Vertical shifting of lens (Case2)

For large values of n , the two ring patterns may produce uniform illumination. The bright ring for wavelength λ_1 will fall on the dark ring for wavelength λ_2 , and conversely.

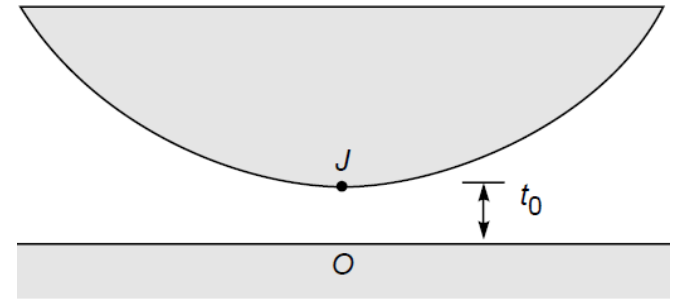
$$2t = n\lambda_1 = (n + 1/2)\lambda_2$$

$$\Rightarrow \frac{2t}{\lambda_1} - \frac{2t}{\lambda_2} = \frac{1}{2} \dots(1)$$

Thus the contrast will be zero, and no fringe pattern will be visible.

Now if we slowly raise the convex lens in the upward direction and let t_0 be such that it satisfies Eq. (1) :

$$t_0 = \frac{\lambda_1 \lambda_2}{4(\lambda_1 - \lambda_2)}$$

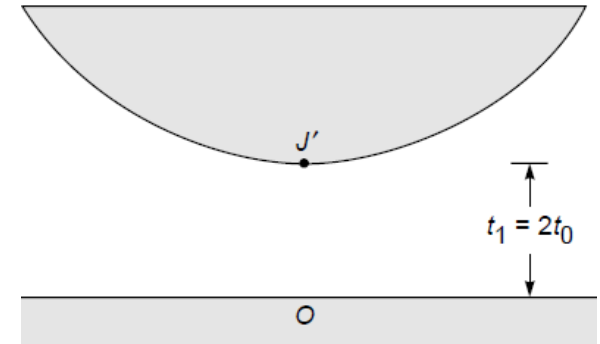


Thus, if point J corresponds to a dark spot for λ_1 , then it will correspond to a bright spot for λ_2 , and conversely. Further, the nearby dark rings for λ_1 will almost fall at the same place as the bright rings for λ_2 , and the interference pattern will be washed out. Thus viewing from a microscope, we will not be able to see any ring pattern.

Newton's Ring : Vertical shifting of lens (Case2)

Now, if the lens is further moved upward by a distance $t_1 = 2t_0$.

$$2t_1 = n\lambda_1 = (n+1)\lambda_2$$
$$\Rightarrow \frac{2t_1}{\lambda_1} - \frac{2t_1}{\lambda_2} = 1 \quad \dots(1)$$



if point J' corresponds to a dark spot for λ_1 , then it will also correspond to a dark spot for λ_2 . The fringe pattern will reappear but now with a slightly weaker contrast.

In this way if we continue to move the lens upward, the fringe system will reappear every time the lens is moved up by a distance

$$2t_0 = \frac{\lambda_1 \lambda_2}{2\Delta\lambda}$$